

1. Final answer:

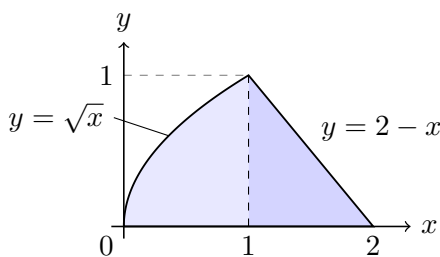
$$\int_0^1 \int_{y^2}^{2-y} f(x, y) dx dy.$$

The two original regions have bounds

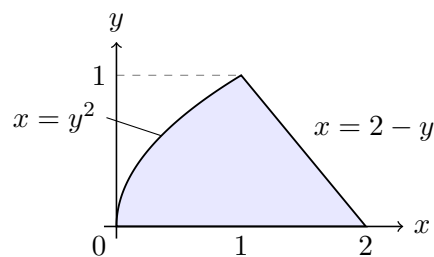
$$D_1 : \begin{cases} 0 \leq x \leq 1, \\ 0 \leq y \leq \sqrt{x} \end{cases} \quad D_2 : \begin{cases} 1 \leq x \leq 2, \\ 0 \leq y \leq 2 - x. \end{cases}$$

They meet along $x = 1$, and the curves meet at $(1, 1)$. A horizontal slice joins $[y^2, 1]$ and $[1, 2 - y]$, giving the combined bounds

$$\begin{cases} 0 \leq y \leq 1, \\ y^2 \leq x \leq 2 - y. \end{cases}$$



Original: two regions split at $x = 1$.



Reversed: one slice, $y^2 \leq x \leq 2 - y$.

As a check, setting $f = 1$ gives area $\frac{2}{3} + \frac{1}{2} = \frac{7}{6}$ in the original order, and $\int_0^1 (2 - y - y^2) dy = \frac{7}{6}$ in the new order.

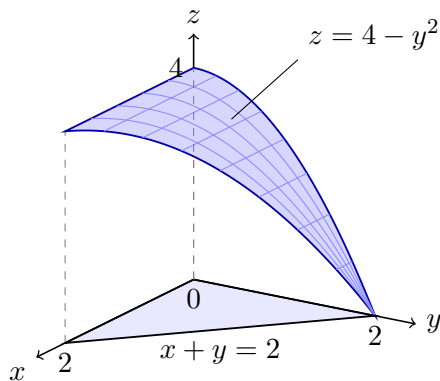
2. Final answer:

$$M = \int_0^2 \int_0^{2-x} \int_0^{4-y^2} (1 + z) dz dy dx.$$

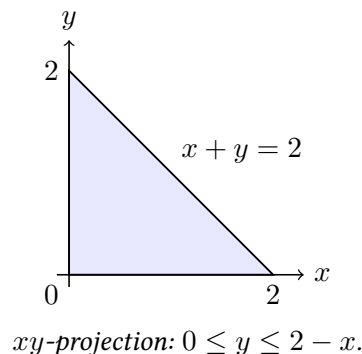
The triangular xy -projection D and the solid E have bounds

$$D : \begin{cases} x \geq 0, \\ y \geq 0, \\ x + y \leq 2 \end{cases} \quad E : \begin{cases} 0 \leq x \leq 2, \\ 0 \leq y \leq 2 - x, \\ 0 \leq z \leq 4 - y^2. \end{cases}$$

Above each base point, the height is $4 - y^2$. Multiplying the volume element by the density $\rho = 1 + z$ gives the mass integral. No evaluation is required.



The surface $(x, y, 4 - y^2)$ above the triangular base.



xy -projection: $0 \leq y \leq 2 - x$.

Using $0 \leq x, y \leq 2$ and $0 \leq z \leq 4$ instead would integrate over a larger rectangular box.

3. **Final answer:**

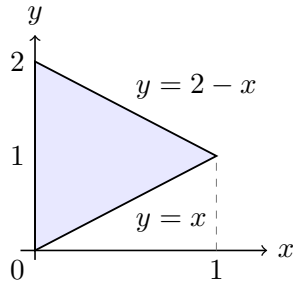
$$\int_0^1 \int_0^y h(x, y) dx dy + \int_1^2 \int_0^{2-y} h(x, y) dx dy.$$

The original bounds are

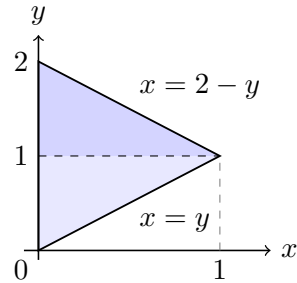
$$\begin{cases} 0 \leq x \leq 1, \\ x \leq y \leq 2 - x. \end{cases}$$

They describe the triangle with vertices $(0, 0)$, $(0, 2)$, and $(1, 1)$. For fixed y , both $x \leq y$ and $x \leq 2 - y$ must hold. Horizontal slices therefore split at $y = 1$, giving

$$\begin{cases} 0 \leq y \leq 1, \\ 0 \leq x \leq y \end{cases} \quad \text{or} \quad \begin{cases} 1 \leq y \leq 2, \\ 0 \leq x \leq 2 - y. \end{cases}$$



Original: $x \leq y \leq 2 - x$.



Reversed: horizontal slices split at $y = 1$.

An equivalent answer is $\int_0^2 \int_0^{\min\{y, 2-y\}} h(x, y) dx dy$; the upper bound can also be written $1 - |y - 1|$.

4. **Final answer:** $\boxed{3}$.

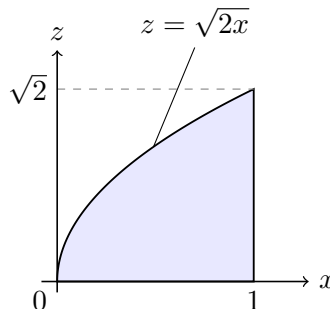
The integration bounds are

$$\begin{cases} 0 \leq y \leq 1, \\ 0 \leq x \leq 1, \\ 0 \leq z \leq \sqrt{2x}. \end{cases}$$

In the z -integration, $g'(y)$ is constant. Since $(\sqrt{2x})^2 = 2x$,

$$\begin{aligned} \int_0^1 \int_0^1 \int_0^{\sqrt{2x}} 2g'(y)z dz dx dy &= \int_0^1 \int_0^1 [g'(y)z^2]_0^{\sqrt{2x}} dx dy \\ &= \int_0^1 g'(y) \left(\int_0^1 2x dx \right) dy = \int_0^1 g'(y) dy \\ &= g(1) - g(0) = 4 - 1 = \boxed{3}. \end{aligned}$$

The x -integral is $[x^2]_0^1 = 1$. The last step uses the Fundamental Theorem of Calculus; no change in the order of integration is needed.



For each $0 \leq y \leq 1$, the same xz -cross section has $0 \leq z \leq \sqrt{2x}$.